Yibing Lu

(646) 371-6139 | yl3498@cornell.edu

yibing-l.github.io | github.com/Yibing-L | home+ code

Education

Cornell University, College of Engineering Jan 2025 - Dec 2025

Master of Engineering in Computer Science | Ithaca, NY

Cornell University, Department of Human Centered Design Aug 2021 - Dec 2024

Bachelor of Science in Design and Environmental Analysis

Minor in Computer Science | Ithaca, NY

Research Experience

Architectural Robotics Lab, Cornell University Feb 2025 - Present

Research Assistant

Advisors: Keith Evan Green, Tapomayukh Bhattacharjee, Haym Hirsh

home+ — Gesture-Controlled Assistive Robotics

- Led development of home+, a gesture-controlled mobile assistive robot for people with upper-limb mobility limitations.
- Integrated gesture recognition and RGB-D perception to translate arm and hand movements into robot actions.
- Developed ROS 2/MoveIt 2 motion planning and encoder-based closed-loop control for coordinated base-and-arm manipulation.
- Prepared study protocols and experimental materials and obtained IRB approval to evaluate gesture interaction under upper-limb movement constraints.
- Led a retrospective of home+'s design evolution (2009–2026), resulting in a first-author manuscript under review at HRI 2027.

Design and Augmented Intelligence Lab, Cornell University Jun 2022 - Dec 2024

Research Assistant | Advisor: Saleh Kalantari

- Built data collection and visualization tools for indoor wayfinding research, including an interface for logging user trajectories and spatial decision points in real time.
- Implemented a trajectory analysis pipeline to extract navigation patterns, hesitation events, and route efficiency metrics from raw position data.
- Supported IRB-approved user studies, contributing to experiment design, participant coordination, and manuscript preparation.

Research Outputs

Under Review

Lu, Y., Hirsh, H., Bhattacharjee, T., Walker, I., & Green, K. E. Seventeen Years in the Making: Tracing Long-Term Design in Assistive Robotic Furnishings. *HRI 2027.* Under review.

Poster and Extended Abstract

Lu, Y., Bhattacharjee, T., Hirsh, H., Green, K. E. home+: Gesture-to-Action Pipeline for an Assistive Robotic Table-Lamp. *Northeast Robotics Colloquium (NERC),* 2025. Poster and extended abstract.

Technical and Research Skills

Programming Languages: Python, MATLAB, C++, Java, SQL, HTML, LaTeX.

Libraries and Computing Tools: PyTorch, TensorFlow, scikit-learn, OpenCV, MediaPipe, pandas, NumPy, Matplotlib, CUDA, Linux/Ubuntu, Git.

Robotics and Perception: ROS, ROS 2, MoveIt 2, behavior trees, motion planning, inverse kinematics, mobile manipulation, RGB-D perception, point-cloud processing, 3D mapping, Arduino, PID control, sensor/actuator integration.

Machine Learning and Algorithms: Deep learning and neural networks, decision trees, search algorithms, classification, regression.

Human-Centered Design and Research: Accessibility-informed design, experimental design, study protocols and IRB preparation, usability testing, data visualization; prototyping with SolidWorks, Rhino, Fusion 360, Figma, Adobe Creative Suite.

Industry Experience

Nanjing Weiyan Network Technology, China Jan 2024 - Aug 2024

Software Engineer Intern

- Developed a web application with ML-based features; contributed to backend architecture and data pipelines.

Nanjing Kaijiang Technology, China May 2022 - Aug 2022

UX Design Intern

- Designed mobile interfaces through user research, prototyping, and usability testing.

Awards

Krouse Fellowship and Human Ecology Undergraduate Research Fund, Cornell (2024)

Summer Undergraduate Research Fellowship, Center for Integrative Developmental Science, Cornell (2023)

Dean's List (6 semesters)